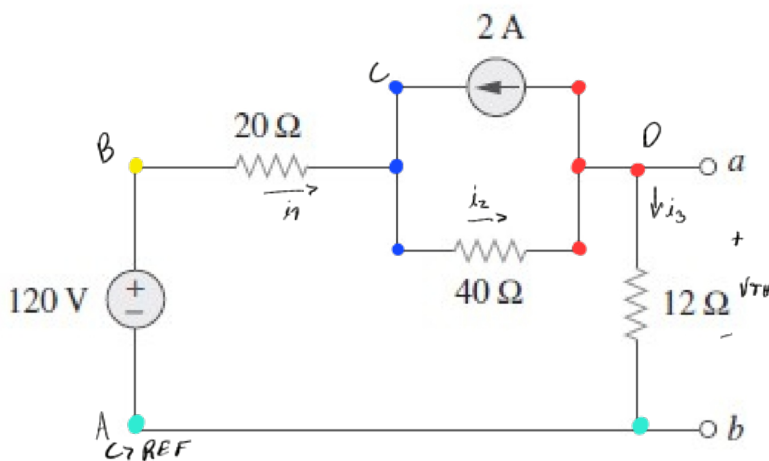


1)



$$V_B = 120V$$

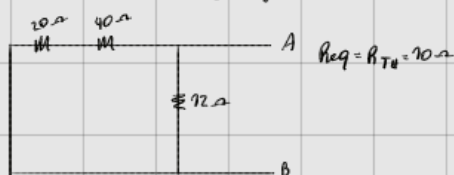
$$\text{Malha C: } 2 + \frac{120 - V_C}{20} = \frac{V_C - V_D}{40} \quad \times 40 \Rightarrow 80 + 240 - 2V_C - V_D = 0 \Rightarrow 3V_C - V_D = 320$$

$$\text{Malha D: } 2 + \frac{V_D}{12} = \frac{V_C - V_D}{40} \quad \times 120 \Rightarrow 960 + 40V_D = 12V_C - 12V_D \Rightarrow 12V_C - 52V_D = 960$$

$$V_C = \frac{980}{9} \quad V_D = \frac{20}{3}$$

$$V_{TH} = V_D = \frac{20}{3} \approx 6,67V$$

* Calcular R_{TH} : Resistor desligar os fontes e calcular o R_{eq}

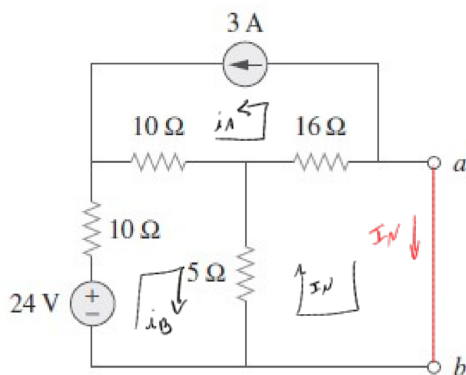


$$* I_N = V = R \cdot I$$

$$I_N = 0,667A$$



2)



$$* i_A = 3A$$

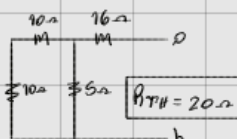
* Malha B:

$$-24 + 10i_B + 10i_B + 30 + 5i_B = 0$$

$$25i_B = -6$$

$$i_B = -\frac{6}{25} A = -0,24A$$

* Calculando o resistor de T_H :



* Com o auto-falado:

$$* i_A = 3A$$

$$* \text{Malha D: } -24 + 10i_B + 30i_A + 10i_B + 5i_B - 5I_N = 0$$

$$25i_B - 5I_N = -6$$

$$i_B = \frac{3}{5} = 0,6A$$

* Malha I_N :

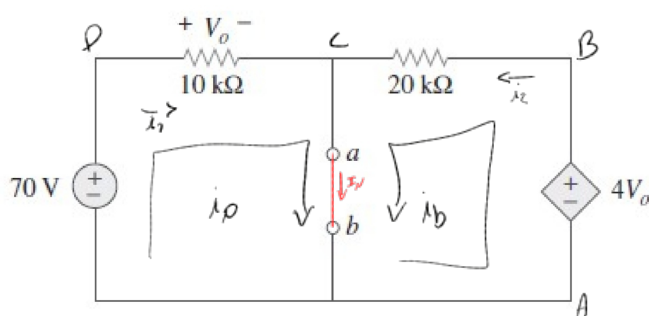
$$10I_N + 16i_B + 5I_N - 5i_B = 0$$

$$-5i_B + 21I_N = -48$$

$$I_N = -2,46A$$

$$* V_{TH} \Rightarrow V_{TH} = 20,246 = 49,2V$$

3)



$$* I_N = i_1 + i_2$$

$$\text{Malha A: } -70 + 10 \cdot 10^3 i_1 = 0 \Rightarrow i_1 = \frac{70}{10^4} = 7 \cdot 10^{-3}$$

$$\therefore I_N = 21 \cdot 10^{-3} A$$

$$\therefore R_{TH} = \frac{60}{21 \cdot 10^{-3}} = 2857,14 \Omega$$

$$\text{Malha B: } -4 \cdot (10 \cdot 10^3 + 10^3) + 20 \cdot 10^3 i_2 = 0 \Rightarrow i_2 = 11 \cdot 10^{-3}$$

* Para calcular o resistor de T_H sem fonte dependente, não pode tirar ele do circuito

$$V_D = 70V \quad i_1 = \frac{70 - V_C}{10^4} \quad i_2 = \frac{V_B - V_C}{20 \cdot 10^3} \quad V = R \cdot i \Rightarrow i = \frac{V}{R}$$

$$V_B = 4V_C$$

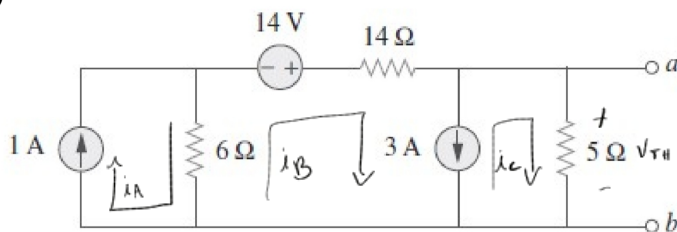
$$V_D = 70 - V_C \Rightarrow V_B = 280 - 4V_C \Rightarrow V_C = 40V$$

$$* \text{Malha C: } \frac{V_B - V_C}{2 \cdot 10^4} + \frac{70 - V_C}{10^4} = 0 \quad (\times 2 \cdot 10^4)$$

$$V_B - V_C + 140 - 2V_C = 0 \Rightarrow V_B - 3V_C + 140 = 0 \Rightarrow 280 - 4V_C + 140 = 0 \Rightarrow V_C = 60V$$

$$V = R \cdot i$$

4)

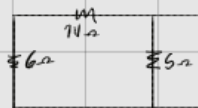


$$i_D = 1A \quad * \text{Fonte de corrente: } i_B - i_D = 3A$$

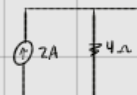
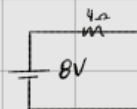
$$* \text{Supermálho: } 6(i_B - 1) - 14 + 14i_B + 5i_C = 0 \Rightarrow 20i_B + 5i_C = 20$$

$$* V_{TH} = 7 \Rightarrow V_{TH} = 5, i_C \Rightarrow V_{TH} = 5,9,6 = 8$$

* Calculando o R_{TH} :



$$R_{eq} = R_{TH} = 4\Omega$$



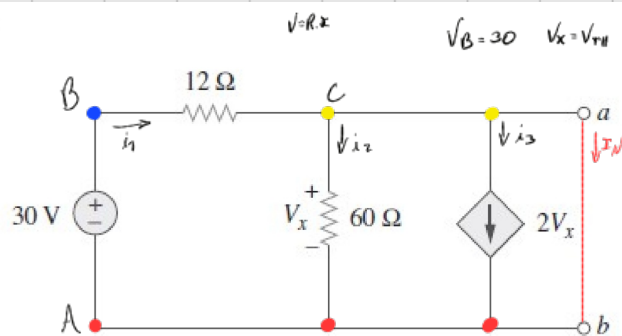
* Calculando o I_N : $I = \frac{V}{R}$

$$I_N = \frac{8}{4} = 2A$$

$$i_B = 1,4A$$

$$i_C = 1,6A$$

5)



$$V_x = 60 i_2 \Rightarrow V_x = V_c$$

$$i_2 = \frac{V_c}{60} \quad i_3 = 2 \cdot V_c$$

* Na c: $i_1 = i_2 + i_3$

$$\frac{30 - V_c}{12} = \frac{V_c}{60} + 2V_c \quad (\times 60)$$

$$150 - 5V_c = V_c + 120V_c$$

$$126V_c = 150 \Rightarrow V_c = \frac{150}{126} \approx 1,19V$$

$$V_{TH} = 1,19V$$

* Quando fechamos os terminais, podemos ver

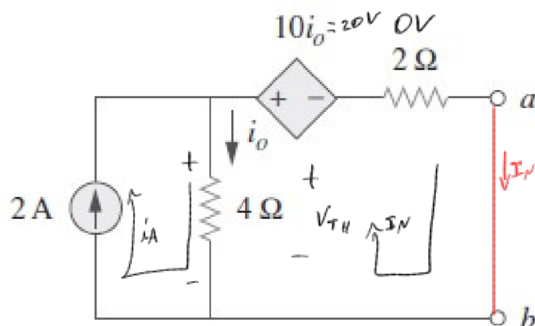
que zeramos o corrente que passa pelo

resistor de 60Ω, portanto, $V_x = 0$.

$$\therefore I_N = \frac{30}{12} = 2,5A \Rightarrow I_N = 2,5A$$

$$* V = R \cdot I \Rightarrow R_{TH} = \frac{1,19}{2,5} = 0,476\Omega \Rightarrow R_{TH} \approx 0,48\Omega$$

6)



* V_{TH} sobre influência do fonte de corrente: $V_{TH} = -20 + 8 = -12 \therefore V_{TH} = 12V$

$$i_D = i_A - I_N \quad i_A = 2A$$

$$i_D = 2 - I_N$$

* Malho I_N :

$$4(I_N - 2) + 10(2 - I_N) + 2I_N = 0$$

$$4I_N - 8 + 20 - 10I_N + 2I_N = 0$$

$$-4I_N = -12$$

$$I_N = 3A$$

$$* V = R \cdot I \Rightarrow R_{TH} = \frac{12}{3} = 4\Omega$$